

Multi-Agent Control Scheme for Power Distribution in a Multi-Three-Phase Motor Drive Fed by a Stacked Polyphase Bridges Converter

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Abstract—A multi-three-phase machine can be seen as a multi-agent system. An agent consists of a three-phase winding set of the multi-three-phase machine, fed by a dedicated three-phase power electronic converter, and steered by means of a dedicated three-phase controller. The series connection of all the three-phase power electronic converters to a single dc power source creates a so-called stacked polyphase bridges converter, and allows for the use of power electronic components with low voltage rating. A major advantage of a multi-three-phase drive is its power sharing capability: not all the agents must contribute equally to the torque demand. A major disadvantage of a stacked polyphase bridges converter, on the other hand, is its inherent instability in motoring mode. The goal of this paper is therefore to propose a multi-agent control strategy that exploits the power sharing capability of the multi-three-phase drive, and simultaneously ensures a stable distribution of the total dc-bus voltage over the agents, even if the agents are not identical. The multi-agent control strategy is in complete accordance with the modularity of the hardware: only local measurements, local computations, and neighbor-to-neighbor communication are required, without the interference of a central or master controller. The control strategy is validated on a 4 kW multi-three-phase axial-flux PMSM working in motoring mode, by means of both simulations and experiments.

Index Terms—Consensus, distributed control, multi-three-phase motor drive, multi-agent system, series dc-link connection.

I. INTRODUCTION

MULTI-THREE-PHASE machines, also called multiple three-phase machines, combine the modularity and redundancy of modular motor drives with the maturity level of

three-phase technology [1], [2], [3]. Such a multi-three-phase machine can be split into multiple three-phase winding sets with separated neutral points, each fed by their own dedicated, well-consolidated three-phase two-level voltage source inverter (2L-VSI).

A multi-three-phase machine can be controlled by means of one central controller by making use of vector space decomposition [1], [2], [4]. This technique projects the fundamental, harmonic and zero-sequence components of the stator currents to mutually orthogonal subspaces, which gives the opportunity to control not only the fundamental, but also the harmonic current components. However, the complexity of the vector space decomposition method increases with the number of phases. Furthermore, the central controller itself is a single point of failure. The vector space decomposition method is hence in conflict with the scalability and the redundancy of the multi-three-phase machine itself.

Another possibility is to equip each of the three-phase sets with its own dedicated three-phase controller, which allows to control the fundamental stator current component only. This strategy is called the multi- dq modeling method, or multi-stator modeling approach [5], [6], [7]. The authors of [8] propose a methodology based on a multi-variable frequency analysis to take the magnetic coupling between the three-phase winding sets into account in the tuning procedure of the PI parameters of the decentralized controllers. This strategy makes it possible to apply conventional three-phase control to the multiple three-phase winding sets, and to provide the same level of modularity, redundancy and scalability to the control as to the hardware.

The combination of a three-phase winding set, its three-phase 2L-VSI, and its dedicated three-phase controller can hence be regarded as an agent in a multi-agent system [9], [10], [11]. The agents are true peers and can work completely independent of one another. A communication network between the agents also allows them to work towards a common goal. Such a common goal can, for instance, be to ensure that the multi-three-phase machine always delivers the demanded torque. A useful property of a multi-three-phase machine is that the agents do not need to contribute equally to this torque demand (which is equivalent to a certain power demand under constant speed) [5], [12], [13]. In [9], [10], a multi-agent control strategy based on static average consensus is proposed that allows the agents to compute so-called sharing coefficients online, in a completely

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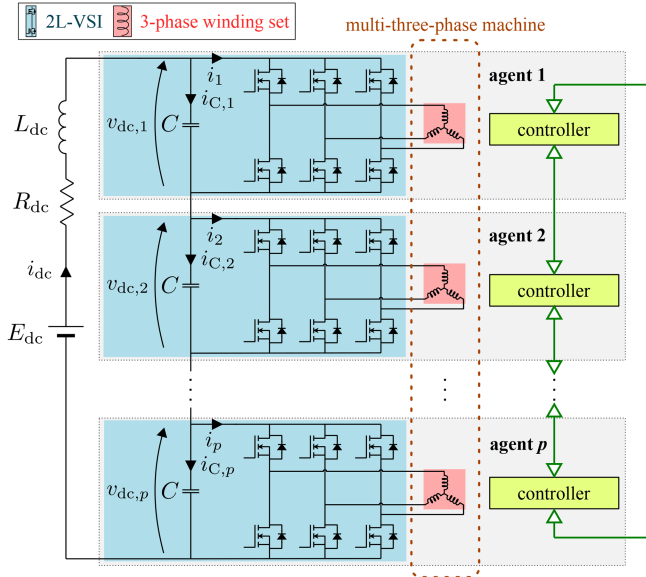


Fig. 1. The stacked polyphase bridges converter consists of a series connection of multiple 2L-VSIs. Each 2L-VSI feeds a three-phase winding set of a multi-three-phase machine, and is controlled with a dedicated three-phase controller. The combination of a three-phase winding set, its 2L-VSI, and its controller is called an agent. Neighbor-to-neighbor communication allows the agents to work together to reach a common goal.

decentralized way. An agent with a low sharing coefficient conducts less current than an agent with a high sharing coefficient, and thus contributes less to the total torque demand. This feature proves, for instance, useful when the agents are not identical by design, or when there are imbalances in the machine windings or the 2L-VSIs (e.g. due to production tolerances, or aging). However, the method for decentralized torque distribution described in [9], [10] is only valid for the case in which all the agents are connected in parallel to the same dc power source, or fed by multiple independent dc power sources.

In this paper, the agents are connected in series to the same dc power source, and together they form a so-called stacked polyphase bridges (SPB) converter [14], as illustrated in Fig. 1. This configuration constitutes the advantage that the power semiconductor switches and capacitors can have voltage ratings lower than the full dc-link voltage E_{dc} , without the need for a step-down converter. This feature proves beneficial for medium to high voltage and power applications (e.g. offshore wind turbines [15]), and automotive applications (e.g. electric vehicles [16], [17]). The main disadvantage of the SPB converter, however, is that the series connection is only inherently stable in generating mode, but not in motoring mode [18]. An active voltage balancing algorithm is required to make sure that the breakdown voltages of the power semiconductor switches and capacitors are not exceeded in motoring mode [19]. The multi-agent voltage balancing algorithm presented in [11] is able to keep the dc-link capacitor voltages of the agents stable by making use of local measurements, local computations, and local agent interactions only. A short-coming of this multi-agent voltage balancer, however, is that it only allows the agents to contribute equally to the torque. The uneven power sharing

capability of a multi-three-phase drive, which can lead to a loss reduction or even a prolonged lifetime when the agents differ from each other, is hence lost.

The goal and main contribution of this paper is therefore to provide a decentralized, multi-agent voltage balancing algorithm for an SPB converter in motoring mode that preserves the uneven power distribution capability of a multi-three-phase machine. It makes use of static [9], [10], [20] and dynamic [11], [21], [22] average consensus to determine respectively the sharing coefficients and the dc-link capacitor voltage references for the agents. The main benefits of the proposed algorithm are:

- the dc-bus voltages of the agents remain stable in motoring mode;
- the agents can take their present condition into account in the online calculation of their sharing coefficients, and hence do not need to be identical;
- no central controller is required: the algorithm is solely based on local computations in the dedicated controllers of the agents;
- there is no need for a central voltage sensor to measure the total dc-bus voltage E_{dc} : the computations of the agents are based on local measurements;
- only neighbor-to-neighbor communication is required;
- the algorithm is easily scalable: changing the number of agents in the series connection does not lead to a complete redesign of the control scheme.

An overview of the multi-three-phase machine and SPB converter topology used throughout this paper is given in Section II. Section III covers the relations between power, voltage and current in an SPB converter, which are required for a good understanding of the proposed multi-agent control strategy presented in Section IV. Finally, the proposed algorithm is demonstrated and validated for the case of a fifteen-phase, surface-mounted axial-flux permanent magnet synchronous machine (AFPSM), split into five three-phase agents in Section V.

II. MOTOR DRIVE ARCHITECTURE

A multi-three-phase machine consists of p three-phase winding sets, each with their own separate neutral point, and with a phase shift of $2\pi/3$ electrical radians between their phases. In this work, each three-phase winding set is fed by means of a standard three-phase 2L-VSI, with its own dc-link capacitor C . The p 2L-VSIs are connected in series to a single dc power supply, which provides the total dc-bus voltage E_{dc} . This series connection of 2L-VSIs is called a stacked polyphase bridges converter, and is illustrated in Fig. 1.

Each three-phase winding set and its corresponding 2L-VSI is steered by means of a dedicated three-phase controller. The controller receives feedback on its three phase currents and its local dc-link capacitor voltage $v_{dc,x}$ ($x \in \{1, \dots, p\}$) from respectively three local current sensors and one local voltage sensor. The combination of a three-phase winding set, its 2L-VSI, its controller, and its sensors is called an agent. In this work, neighboring agents have a phase shift of $2\pi/(3p)$ electrical radians between them.

A ring communication network enables neighbor-to-neighbor communication between the agents. This ring network makes the drive easily scalable, and provides a good trade-off between the number of required communication links, and the requirement that each agent must be in contact with at least one other agent in order for the consensus algorithms discussed in Section IV to converge.

III. RELATION BETWEEN POWER, VOLTAGE AND CURRENT IN AN SPB CONVERTER

For the SPB converter topology, the capacitor voltages $v_{dc,x}$ over the p agents are related as follows:

$$L_{dc} \frac{di_{dc}}{dt} = E_{dc} - R_{dc} i_{dc} - \sum_{x=1}^p v_{dc,x}, \quad (1)$$

with L_{dc} (H), i_{dc} (A), E_{dc} (V), and R_{dc} (Ω) the inductance, current, voltage, and resistance of the voltage source, respectively, as indicated in Fig. 1. The variation in the dc-link capacitor voltage of agent x depends on its capacitance C (F), the dc-link current i_{dc} , and the current i_x to its 2L-VSI:

$$C \frac{dv_{dc,x}}{dt} = i_{dc} - i_x \quad (\forall x \in \{1, \dots, p\}). \quad (2)$$

If the inverter-, stator-, and mechanical losses of the agents are negligible, the power P_x that goes into agent x can be calculated as:

$$P_x = v_{dc,x} i_x, \quad (3)$$

or as:

$$P_x = t_x \Omega, \quad (4)$$

with Ω (rad/s) the mechanical speed, and

$$t_x = \frac{3}{2} N_p [\Psi_m i_{q,x} + (L_{d,x} - L_{q,x}) i_{d,x} i_{q,x}] \quad (5)$$

the torque (Nm) provided by agent x , with N_p the number of pole pairs, Ψ_m (Wb) the flux linkage caused by the permanent magnets, $i_{d,x}$ and $i_{q,x}$ (A) the dq -current components of agent x , and $L_{d,x}$ and $L_{q,x}$ (H) the stator inductances. As the speed Ω is the same for all the agents in the electrical machine, the distribution of a total power demand $P_{ref} = \sum_x P_x$ among the agents is equivalent to the division of a total torque reference $T_{ref} = p t_{ref}$ into p agent-specific torque set-points $t_{ref,x} = \sigma_x t_{ref}$, satisfying:

$$\sum_{x=1}^p \sigma_x = p \quad \text{or} \quad \sum_{x=1}^p t_{ref,x} = T_{ref}. \quad (6)$$

The general torque reference per agent $t_{ref} = T_{ref}/p$ is coming from the user, and is known by every agent x . The agent-specific current references $i_{d,ref,x}$ and $i_{q,ref,x}$ can be computed from the agent-specific torque references $t_{ref,x}$ by means of, for instance, a maximum-torque per ampere control method [23]. For the specific case of a PMSM with surface-mounted permanent magnets without flux-weakening, $i_{d,ref,x}$ is set equal to zero for all the agents, while $i_{q,ref,x}$ is proportional to the torque reference, i.e. $i_{q,ref,x} = (2t_{ref,x})/(3N_p \Psi_m)$.

The sharing coefficient σ_x of agent x hence indicates the portion of the total power demand that agent x will provide. When all the agents are identical and can contribute equally to the total power, the sharing coefficients σ_x are equal to one for all the agents. When the agents are not identical (due to production tolerances, different types of power electronic converters, etc.), the sharing coefficients can differ between the agents. Large variations among the agents result in large variations among the σ_x .

When the power is not distributed evenly among the agents (i.e. when not all σ_x are equal to one), neither will the total dc-bus voltage E_{dc} be distributed evenly among the agents. From (1)–(6), the following relations hold for the voltage set-point $v_{dc,x}^*$ that needs to accompany the torque reference $t_{ref,x} = \sigma_x T_{ref}/p$ in steady-state:

$$v_{dc,x}^* = \sigma_x \frac{T_{ref} \Omega}{p i_{dc}} = \sigma_x \sum_{x=1}^p \frac{v_{dc,x}}{p} \approx \sigma_x \frac{E_{dc}}{p}. \quad (7)$$

The sharing coefficients σ_x of the agents hence also indicate how the total dc-bus voltage E_{dc} is distributed among them.

IV. MULTI-AGENT CONTROL SCHEME

The complete multi-agent control strategy is schematized in Fig. 2. Four main parts can be distinguished:

- A deterministic static average consensus algorithm (see Section IV-A) is used to distribute the power among the agents. The algorithm determines the sharing coefficients σ_x , based on the present condition of each agent x . These sharing coefficients immediately allow to determine a torque reference for each agent x , i.e. $t_{ref,x} = \sigma_x t_{ref}$. The current references $i_{d,ref,x}$ and $i_{q,ref,x}$ can then be derived from this torque reference.
- A dynamic average consensus algorithm (see Section IV-B) determines the average dc-link capacitor voltage $\bar{v}_{dc,x} \approx \sum_y v_{dc,y}/p$ in a decentralized way. In combination with the sharing coefficients σ_x coming from the deterministic static average consensus algorithm, the agent-specific capacitor voltage set-points $v_{dc,x}^* = \sigma_x \bar{v}_{dc,x}$ can be determined.
- The active voltage balancer of Section IV-C regulates the dc-link capacitor voltages $v_{dc,x}$ towards their set-points $v_{dc,x}^*$ by adding small increments to the initial agent-specific current references $i_{d,ref,x}$ and $i_{q,ref,x}$.
- The decentralized current control algorithm of Section IV-D regulates the current components of the agents towards their current set-points $i_{d,x}^*$ and $i_{q,x}^*$ coming from the active voltage balancer.

These four main parts are elaborated upon in the following subsections.

A. Deterministic Static Average Consensus for Power Distribution

The goal of the deterministic static average consensus algorithm is to let the agents decide by themselves how the total torque demand T_{ref} will be distributed among them, without the interference of a central or master controller. The distribution

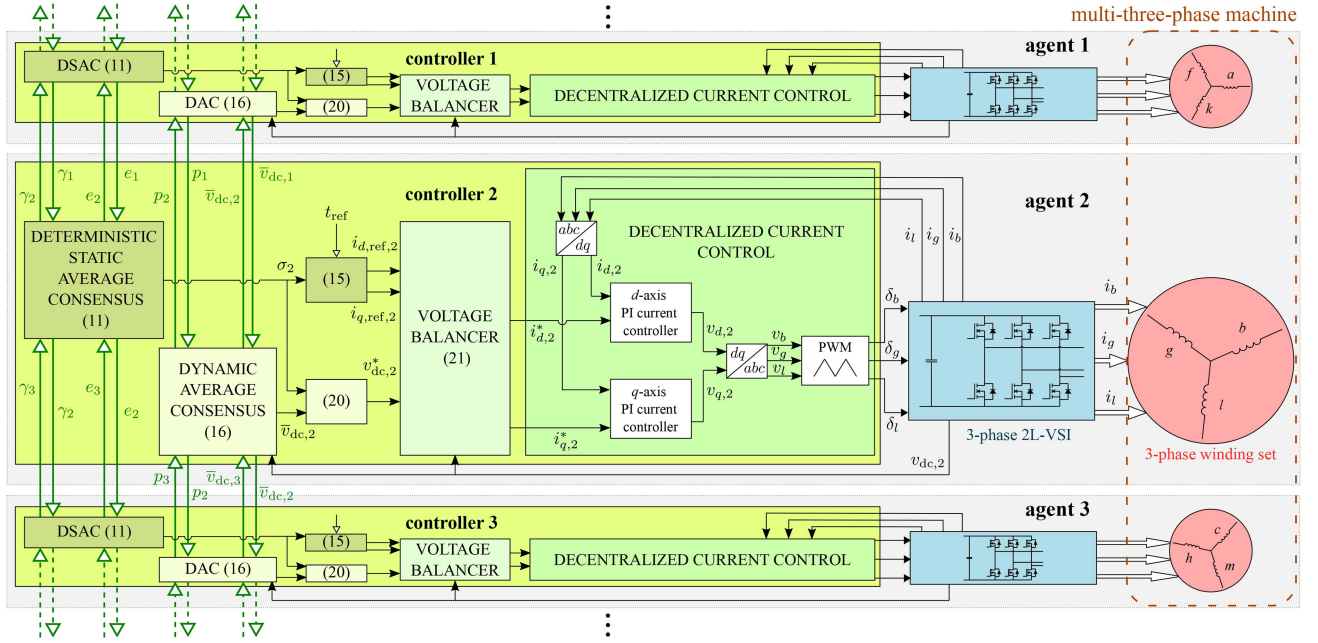


Fig. 2. Detailed control scheme of agent 2. Each agent exchanges its parameters e_x and γ_x with its two closest neighbors (which are agent 1 and 3 for agent 2), in order to calculate its sharing coefficient σ_x by means of a deterministic static average consensus algorithm (DSAC, see Section IV-A). This sharing coefficient allows to determine the agent-specific current component references $i_{d,\text{ref},x}$ and $i_{q,\text{ref},x}$ from the general torque set-point t_{ref} per agent. The exchange of the parameters p_x and $\bar{v}_{dc,x}$ with the two closest neighbors, in combination with the measurement of the dc-link capacitor voltage $v_{dc,x}$, allows to conduct a dynamic average consensus algorithm (DAC, see Section IV-B) in order to estimate the average dc-link capacitor voltage $\bar{v}_{dc,x} \approx E_{dc}/p$ of all the agents. From σ_x and $\bar{v}_{dc,x}$, the agent-specific voltage reference $v_{dc,x}^*$ can be computed. The voltage balancer (see Section IV-C) then adapts the current component references based on the difference between the measured dc-link capacitor voltage $v_{dc,x}$ and its reference $v_{dc,x}^*$. Finally, the decentralized PI current controller of agent x (see Section IV-D) brings the current components $i_{d,x}$ and $i_{q,x}$ to their final set-points $i_{d,x}^*$ and $i_{q,x}^*$ by determining the duty cycles to steer the switches of the 2L-VSI of agent x based on feedback of the three phase currents of agent x and the rotor position θ .

process must take into account the differences between the agents and the present condition of each agent, without a central or master controller that keeps an overview of the status of all the agents. For this purpose, each agent x is equipped with its own quadratic cost function $\mathcal{C}_x$:

$$\mathcal{C}_x = a_x \sigma_x^2 + b_x \sigma_x + c_x, \quad (8)$$

with $a_x > 0$, $b_x \geq 0$ and $c_x \geq 0$ agent-specific parameters which reflect the state of each individual agent [20], [24]. The use of the incremental cost γ_x , defined as:

$$\gamma_x = \frac{d\mathcal{C}_x}{d\sigma_x} = 2a_x \sigma_x + b_x, \quad (9)$$

makes it possible to mathematically express when the torque distribution is fair: when the incremental costs of all the agents are equal, i.e. when

$$\gamma_1 = \gamma_2 = \dots = \gamma_x = \dots = \gamma_p = \gamma^* = \frac{p + \sum_{x=1}^p \frac{b_x}{2a_x}}{\sum_{x=1}^p \frac{1}{2a_x}}, \quad (10)$$

it is equally (dis)advantageous for each agent x to change its sharing coefficient σ_x , or, in other words, to change its torque contribution.

From a mathematical point of view, the goal of the deterministic static average consensus algorithm is hence to let the agents reach agreement on a common value for their incremental cost γ_x , while simultaneously ensuring that constraint (6) is satisfied, i.e. that the total torque demand is delivered. To reach

this goal, the following deterministic static average consensus algorithm (inspired on [10], [20], [24]) is conducted by every agent x , at every discrete time instant k (with an update period $T_c = 1/f_c$ between subsequent time instants):

$$\begin{aligned} \gamma_x(k+1) &= \frac{1}{3}\gamma_{x-1}(k) + \frac{1}{3}\gamma_x(k) + \frac{1}{3}\gamma_{x+1}(k) \\ &\quad + \varepsilon e_x(k); \end{aligned} \quad (11a)$$

$$\sigma_x(k+1) = f_x(\gamma_x(k+1)); \quad (11b)$$

$$\begin{aligned} e_x(k+1) &= \frac{1}{3}e_{x-1}(k) + \frac{1}{3}e_x(k) + \frac{1}{3}e_{x+1}(k) \\ &\quad - [\sigma_x(k+1) - \sigma_x(k)]. \end{aligned} \quad (11c)$$

The state variables are initialized as follows:

$$\begin{cases} \gamma_x(0) = b_x; \\ \sigma_x(0) = 0; \\ e_x(0) = 1. \end{cases} \quad (12)$$

Agent x hence only needs information from itself, and from its two neighboring agents $x-1$ and $x+1$. The parameter e_x is the local estimate of agent x of the discrepancy between the actual sum of the sharing coefficients $\sum_{x=1}^p \sigma_x$ and its required

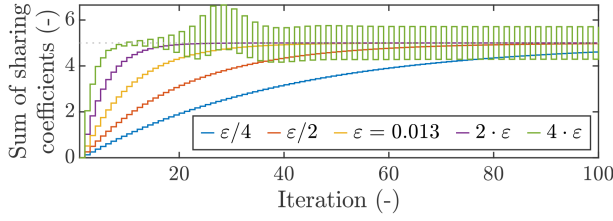


Fig. 3. Effect of the learning gain ε ($p = 5$).

value p . The function $f_x(\gamma_x)$ is defined as:

$$f_x(\gamma_x) = \begin{cases} \sigma_{\min}, & \text{if } \gamma_x < \gamma_{\min,x}; \\ \frac{\gamma_x - b_x}{2a_x}, & \text{if } \gamma_{\min,x} \leq \gamma_x \leq \gamma_{\max,x}; \\ \sigma_{\max}, & \text{if } \gamma_x > \gamma_{\max,x}. \end{cases} \quad (13)$$

The constraints $\gamma_{\min,x} = 2a_x\sigma_{\min} + b_x$ and $\gamma_{\max,x} = 2a_x\sigma_{\max} + b_x$ on the incremental cost are determined by the constraints $\sigma_{\min}$ and $\sigma_{\max}$ on the sharing coefficients. $\sigma_{\min}$ and $\sigma_{\max}$ are in their turn derived from the current- and voltage constraints:

$$\sqrt{i_{d,\text{ref},x}^2 + i_{q,\text{ref},x}^2} \leq i_{\max}, \quad (14a)$$

$$t_x > 0, \quad (14b)$$

$$v_{dc,\min} \leq \sigma_x E_{dc}/p \leq v_{dc,\max}. \quad (14c)$$

The learning gain ε is a sufficiently small positive constant, which can be used to fine-tune the convergence speed of algorithm (11), as explained in [10] and illustrated in Fig. 3. A very small ε results in slow convergence of $\sum_{x=1}^p \sigma_x$ to p . A large ε , on the other hand, can lead to instability.

Eventually, the agent-specific torque references $t_{\text{ref},x}$ ($\forall x \in \{1, \dots, p\}$) are found by combining the sharing coefficients σ_x with the global torque reference $t_{\text{ref}} = T_{\text{ref}}/p$ which is known by each agent x :

$$t_{\text{ref},x}(k) = \sigma_x(k)t_{\text{ref}}(k). \quad (15)$$

The agent-specific current references $i_{d,\text{ref},x}(k)$ and $i_{q,\text{ref},x}(k)$ can then be found by means of a maximum-torque per ampere method, for instance [23].

B. Dynamic Average Consensus for Voltage Set-Point Generation

According to (7), the voltage set-point $v_{dc,x}^*$ for agent x depends on its sharing coefficient σ_x , and the time-varying average of all the local dc-link capacitor voltages $v_{dc,\text{av}} = \sum_{y=1}^p v_{dc,y}/p \approx E_{dc}/p$. However, as there is no central or master controller, no central sensor to measure E_{dc} , and as only neighbor-to-neighbor communication is allowed, this average voltage $v_{dc,\text{av}}$ is not readily available. Instead, the dynamic average consensus algorithm of [11], [21], [22] is used to let each agent track this average voltage $v_{dc,\text{av}}$ in a completely decentralized way, based on neighbor-to-neighbor interactions,

local measurements, and local calculations only:

$$q_x(l+1) = \rho q_x(l) + k_p \{ [p_x(l) + \bar{v}_{dc,x}(l)] - \frac{1}{2} [p_{x-1}(l) + \bar{v}_{dc,x-1}(l) + p_{x+1}(l) + \bar{v}_{dc,x+1}(l)] \} \quad (16a)$$

$$p_x(l+1) = p_x(l) + k_i \{ \bar{v}_{dc,x}(l) - \frac{1}{2} \bar{v}_{dc,x-1}(l) - \frac{1}{2} \bar{v}_{dc,x+1}(l) \} \quad (16b)$$

$$\bar{v}_{dc,x}(l+1) = v_{dc,x,f}(l) - q_x(l+1) \quad (16c)$$

with

$$v_{dc,x,f}(l) = \alpha v_{dc,x}(l) + [1 - \alpha] v_{dc,x,f}(l-1). \quad (17)$$

The update period between two subsequent discrete time instants l and $l+1$ is equal to $T_{vb} = 1/f_{vb}$. q_x and p_x are internal state variables of the consensus algorithm, while $\bar{v}_{dc,x}$ is the estimate of agent x of the time varying average dc-link capacitor voltage $v_{dc,\text{av}} = \sum_{y=1}^p v_{dc,y}/p$. Analytical expressions for the parameters ρ , k_p and k_i that result in the fastest convergence of algorithm (16) are derived in [21] and summarized in [11], [22]:

$$\rho = \begin{cases} \rho_0, & 0 < \lambda_r \leq 3 - \sqrt{5} \\ \rho_1, & 3 - \sqrt{5} < \lambda_r \leq 1 \end{cases} \quad (18a)$$

$$k_p = [\rho(1 - \rho)\lambda_r] / [\lambda_n(\rho + \lambda_r - 1)], \quad (18b)$$

$$k_i = (1 - \rho)/\lambda_2 \quad (18c)$$

with

$$\lambda_r = \lambda_2/\lambda_p, \quad (19a)$$

$$\rho_0 = [8 - 8\lambda_r + \lambda_r^2] / [8 - \lambda_r^2], \quad (19b)$$

$$\rho_1 = \frac{\sqrt{(1 - \lambda_r)(4 + \lambda_r^2(5 - \lambda_r))} - \lambda_r(1 - \lambda_r)}{2(1 + \lambda_r^2)}, \quad (19c)$$

and λ_i ($i \in \{1, \dots, p\}$) the eigenvalues of the Laplacian matrix $\mathbf{L}$, with $\lambda_1 = 0$ and $\Re(\lambda_i) \leq \Re(\lambda_j)$ for $i < j$. When communication is restricted to neighbor-to-neighbor communication, this Laplacian matrix has the following form:

- its diagonal elements l_{xx} ($x \in \{1, \dots, p\}$) are equal to 1,
- its elements l_{xy} ($x \in \{1, \dots, p\}$, and $y = x \pm 1 \in \{1, \dots, p\}$) are equal to $-1/2$,
- its other elements are zero.

The term $k_i \{ \bar{v}_{dc,x}(l) - \frac{1}{2} \bar{v}_{dc,x-1}(l) - \frac{1}{2} \bar{v}_{dc,x+1}(l) \}$ in (16b) can be rewritten as $\frac{1}{2} k_i \{ [\bar{v}_{dc,x}(l) - \bar{v}_{dc,x-1}(l)] + [\bar{v}_{dc,x}(l) - \bar{v}_{dc,x+1}(l)] \}$. In this way, it becomes apparent that this term calculates the average deviation of $\bar{v}_{dc,x}$ from the communicated average voltage estimates $\bar{v}_{dc,x-1}$ and $\bar{v}_{dc,x+1}$. The internal state variable p_x hence represents an initial estimate of the deviation of $v_{dc,x}$ from the average dc-link capacitor voltage $v_{dc,\text{av}}$. The term $k_p \{ [p_x(l) + \bar{v}_{dc,x}(l)] - \frac{1}{2} [p_{x-1}(l) + \bar{v}_{dc,x-1}(l) + p_{x+1}(l) + \bar{v}_{dc,x+1}(l)] \}$ in (16a) can also be rewritten as $\frac{1}{2} k_p \{ [p_x(l) - p_{x-1}(l)] + [p_x(l) - p_{x+1}(l)] + [\bar{v}_{dc,x}(l) - \bar{v}_{dc,x-1}(l)] + [\bar{v}_{dc,x}(l) - \bar{v}_{dc,x+1}(l)] \}$ in order to emphasize that the internal variable q_x improves this initial estimate of the deviation of $v_{dc,x}$ from $v_{dc,\text{av}}$. Whereas p_x only involves the variables $\bar{v}_{dc,y}(l)$ from the

agents $y \in \{x-1, x, x+1\}$ directly, q_x also involves the variables $p_y(l-1)$ ($y \in \{x-1, x, x+1\}$) and the variables $\bar{v}_{dc,z}(l-1)$ ($z \in \{x-2, x-1, x, x+1, x+2\}$) directly (through the communicated $p_y(l)$). Hence, the internal state variable q_x has better knowledge of the past, and involves information about more agents than internal state variable p_x . As a result, (16a) makes the dynamic average consensus algorithm robust against agent dropout. Therefore q_x is used as the estimate of the deviation between $v_{dc,x}$ and $v_{dc,av}$, which is subtracted from the voltage measurement of agent x in (16c), in order to obtain an estimate $\bar{v}_{dc,x}$ of the average dc-link voltage $v_{dc,av}$. The filtering of the voltage sensor output $v_{dc,x}$ with a filtering constant α according to (17) avoids that the voltage average $\bar{v}_{dc,x}$ is heavily influenced by measurement noise, and according to [18], it enhances the stability of the voltage balancer discussed in Section IV-C.

Eventually, the voltage set-points $v_{dc,x}^*$ for the voltage balancing algorithm are found by combining the outputs σ_x of the deterministic static average consensus algorithm, and the outputs $\bar{v}_{dc,x}$ of the dynamic average consensus algorithm:

$$v_{dc,x}^*(l) = \sigma_x(l) \bar{v}_{dc,x}(l). \quad (20)$$

C. Active Voltage Balancing

In generating mode, the dc-link capacitor voltages are inherently stable [18]. In motoring mode, however, the dc-link capacitor voltages $v_{dc,x}$ need to be stabilized and regulated towards their set-points $v_{dc,x}^*$ ($x \in \{1, \dots, p\}$) by means of the active voltage balancer proposed in [11], [18]. At each update period T_{vb} , a stabilization term with a gain g is added to the agent-specific current component references $i_{d,ref,x}$ and $i_{q,ref,x}$:

$$i_{d,x}^*(l) = i_{d,ref,x}(l) + g i_{d,ref}(l) [v_{dc,x}(l) - v_{dc,x}^*(l)], \quad (21a)$$

$$i_{q,x}^*(l) = i_{q,ref,x}(l) + g i_{q,ref}(l) [v_{dc,x}(l) - v_{dc,x}^*(l)]. \quad (21b)$$

When the voltage $v_{dc,x}$ is larger than its set-point $v_{dc,x}^*$, a larger inverter current i_x will be drawn into agent x , according to (3)–(5). A sufficient increase in i_x allows the capacitor voltage to decrease, according to (2). The inverse happens when $v_{dc,x}$ is smaller than its set-point $v_{dc,x}^*$. The current set-points are limited by the current constraints, i.e. $\sqrt{i_{d,x}^{*2} + i_{q,x}^{*2}} \leq i_{max}$, and $t_x > 0$ to ensure motoring mode.

The tuning of the gain g is a trade-off between ripple in the current set-points on the one hand, and deviations of the capacitor voltages from their set-points on the other hand. A high g keeps the dc-link capacitor voltages close to their set-points, but causes a large ripple in the current set-points. A smaller g results in a smaller ripple amplitude in the current set-points, but larger deviations between the dc-link capacitor voltages and their set-points. These effects are demonstrated in Fig. 4 for five agents with identical voltage set-points $v_{dc,x}^* = 48$ V.

D. Decentralized Current Control

Finally, the current components $i_{d,x}$ and $i_{q,x}$ of each agent x are regulated towards their set-points $i_{d,x}^*$ and $i_{q,x}^*$. For this

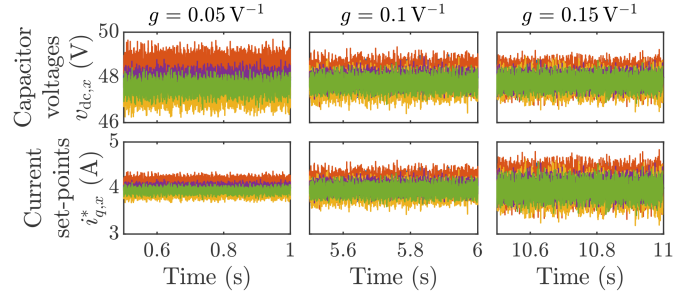


Fig. 4. Measured effect of voltage balancing gain g for five agents with identical voltage set-points $v_{dc,x}^* = 48$ V ($f_{vb} = 10$ kHz, $\Omega = 42$ rad/s).

purpose, p independent three-phase vector PI controllers with sampling and switching frequency f_s are used, as indicated in Fig. 2. However, the agents are still magnetically coupled. A robust, decoupled current control of the agents hence requires a PI parameter tuning procedure that takes this magnetic coupling into account. The state-of-the-art procedure of [8] is followed in this work. The candidate PI parameters K_p and K_i are evaluated by means of a multi-variable frequency analysis, which assesses the effect of a q -current set-point amplitude change in agent x on the current components $i_{d,y}$ and $i_{q,y}$ of the other agents y . For this purpose, the Bode diagrams of the closed-loop transfer functions need to be analyzed. These closed-loop transfer functions can be expressed by means of the following matrix:

$$\frac{I_{dq}}{I_{dq}^*} = \mathbf{K} \left(\mathbf{R}_{dq} + s \mathbf{L}_{dq} - \omega \mathbf{P}_{dq} \left(\mathbf{L}_{dq} - \hat{\mathbf{L}}_{dq} \right) + \mathbf{K} \right)^{-1} \quad (22)$$

with $\mathbf{K} = \text{diag}(K_p + \frac{K_i}{s}, \dots, K_p + \frac{K_i}{s})$, $\mathbf{R}_{dq}$ the diagonal stator resistance matrix (with the stator resistance R_s on all the diagonal elements), s the Laplace variable, $\mathbf{L}_{dq}$ the non-diagonal inductance matrix which includes the magnetic coupling between the agents, ω the electrical speed, $\hat{\mathbf{L}}_{dq} = \text{diag}(L_{d,1}, L_{q,1}, \dots, L_{d,p}, L_{q,p})$ the decoupled inductance matrix, and $\mathbf{P}_{dq} = \text{diag}(\mathbf{P}, \dots, \mathbf{P})$ a permutation matrix with

$$\mathbf{P} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

V. CASE STUDY

The proposed multi-agent control scheme of Fig. 2 is validated on a fifteen-phase, 4 kW segmented armature torus axial-flux permanent magnet synchronous machine with surface-mounted permanent magnets, split into five three-phase agents. No flux-weakening will be considered, which means that $i_{d,ref,x}$ and $i_{d,x}^*$ are equal to zero, and that $i_{q,ref,x}$ is directly proportional to the torque reference: $i_{q,ref,x} = (2t_{ref,x}) / (3N_p \Psi_m)$. The specifications of the AFPMSM are listed in Table I. Only motoring mode will be considered in the simulations and the experiments, as the SPB converter is inherently stable in generating mode [18]. The control parameters are summarized in Table II. In this work, the PI parameters for the decentralized current controllers are

TABLE I
DRIVE PARAMETERS

Parameter	Value
Rated torque	$T_n = 15 \text{ Nm}$
Rated current	$I_n = 11 \text{ A}$
Rated speed	$N_n = 2500 \text{ rpm}$
Number of pole pairs	$N_p = 8$
Stator winding resistance	$R_s = 65 \text{ m}\Omega$
Stator inductance	$L_d = L_q = 309.95 \mu\text{H}$ $L_{s,s} = 316.00 \mu\text{H}$ $L_{s,s-1} = L_{s,s+1} = 111.00 \mu\text{H}$ $L_{s,s-2} = L_{s,s+2} = 15.50 \mu\text{H}$ $L_{s,s-3} = L_{s,s+3} = 9.52 \mu\text{H}$ $L_{s,s-4} = L_{s,s+4} = 8.27 \mu\text{H}$ $L_{s,s-5} = L_{s,s+5} = 6.05 \mu\text{H}$ $L_{s,s-6} = L_{s,s+6} = 6.07 \mu\text{H}$ $L_{s,s-7} = L_{s,s+7} = 5.31 \mu\text{H}$
Self- and mutual inductances ($s \in \{1, 2, \dots, 15\}$)	
Permanent magnet flux linkage	$\Psi_m = 0.02 \text{ Wb}$
Total dc-bus voltage	$E_{dc} = 240 \text{ V}$
DC-link capacitance per agent	$C = 220 \mu\text{F}$

TABLE II
CONTROL PARAMETERS

Parameter	Value
<i>Deterministic static average consensus</i>	
Update frequency	$f_c = 2 \text{ kHz}$
Learning gain	$\varepsilon = 0.013$
Current limit	$i_{\max} = 10 \text{ A}$
Minimum voltage limit	$v_{dc,\min} = 0 \text{ V}$
Maximum voltage limit	$v_{dc,\max} = 100 \text{ V}$
<i>Dynamic average consensus</i>	
Update frequency of the voltage balancer	$f_{vb} = 1 \text{ kHz}$
Filtering constant	$\alpha = 0.01$
Convergence rate	$\rho = 0.6481$
Proportional gain	$k_p = 1.6022$
Integral gain	$k_i = 0.5093$
<i>Active voltage balancer</i>	
Gain	$g = 0.06 \text{ V}^{-1}$
<i>Decentralized current control</i>	
Sampling and switching frequency	$f_s = 10 \text{ kHz}$
Proportional gain	$K_p = 2$
Integral gain	$K_i = 200$

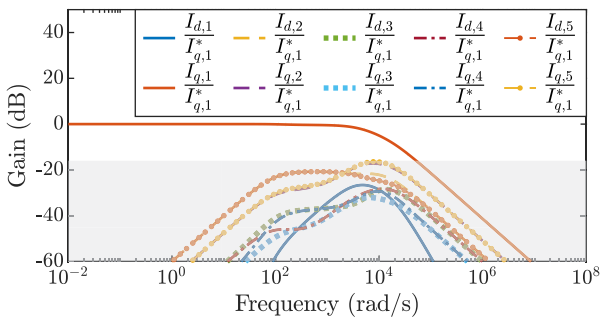


Fig. 5. Bode plot of the closed-loop transfer functions (22) for the candidate PI parameters $K_p = 2$ and $K_i = 200$ at a mechanical speed of 2500 rpm.

tuned in such a way that a step in the q -current set-point of agent x does not deviate the current components $i_{d,y}$ and $i_{q,y}$ of the other agents y with more than 16% of the magnitude of that q -current set-point change. Fig. 5 shows the Bode plots of the closed-loop transfer functions (22).

Although the hardware of the agents is identical (except for unknown production tolerances), the cost functions (8) of the

agents will be chosen differently. It is chosen to set $b_x = c_x = 0$ ($\forall x \in \{1, \dots, p\}$), but to let a_x vary among the agents. This corresponds, for instance, to the situation in which the machine's copper losses can be minimized in case there are differences between the stator resistances $R_{s,x}$ of the agents. It must be noted, however, that the cost coefficients a_x , b_x and c_x are not restricted to this specific example. The coefficients a_x , for instance, could also be used to minimize the conduction losses in the 2L-VSIs by making them proportional to the on-state resistances of the power semiconductor switches. Another option is to make a_x inversely proportional to the stator inductances, in order to obtain a comparable saturation level in all the agents. The coefficients b_x could be used to minimize the switching losses. As the switching losses are proportional to the current running through the power semiconductor switches of the 2L-VSIs, they can be represented by means of the term $b_x \sigma_x$ in cost function (8). As the coefficients c_x are no part of the incremental cost function (9), they will have no effect on the sharing coefficients.

However, in this paper, only the situation $a_x = R_{s,x}$, $b_x = c_x = 0$ will be elaborated upon. The cost coefficients a_x can be varied in a range $[0.73R_s, 1.35R_s]$ around the actual stator winding resistance R_s of each agent. Three situations are examined:

- the situation in which the cost coefficients a_x are equal to the nominal value of the stator resistance R_s for all the agents x ;
- the situation in which the cost coefficients a_x are different for every agent x , but remain constant in time;
- the situation in which the cost coefficients a_x are different for every agent x , and also vary in time.

A. Simulation Results

When all the cost coefficients a_x are equal to the nominal value of the stator resistance R_s , the agents reach agreement on an incremental cost $\gamma^* = 0.13$, as illustrated in the simulation results of Fig. 6. Their sharing coefficients σ_x are all equal to 1. The torque and the dc-bus voltage are hence distributed evenly among the agents, i.e. all the q -current references $i_{q,\text{ref},x}$ are equal to $i_{q,\text{ref}} = (2T_{\text{ref}})/(3pN_p\Psi_m)$, and the voltage set-points $v_{dc,x}^*$ for the voltage balancers are equal to their corresponding outputs $\bar{v}_{dc,x}$ of dynamic average consensus algorithm (16). The dynamic average consensus algorithm outputs a smooth estimate of the average dc-link capacitor voltage for every agent x , i.e. $\bar{v}_{dc,x} \approx \sum_{y=1}^p v_{dc,y}/p \approx E_{dc}/p = 48 \text{ V}$. The individual capacitor voltages $v_{dc,x}$ are correctly regulated towards their voltage set-points. The initial q -current references $i_{q,\text{ref},x}$ are adapted to the q -current set-points $i_{q,x}^*$ by means of the voltage balancers (21) for this purpose. The decentralized current controllers of Section IV-D ensure that the stator current components $i_{d,x}$ and $i_{q,x}$ follow their set-points $i_{d,x}^* = 0 \text{ A}$ and $i_{q,x}^*$ well. The time-varying torque demand T_{ref} is hence met at any time, and the dc-link voltage distribution remains stable, while the mechanical speed N of the drive is varied.

Fig. 7 shows the simulation results for the case in which the cost coefficients a_x are different for every agent x , but

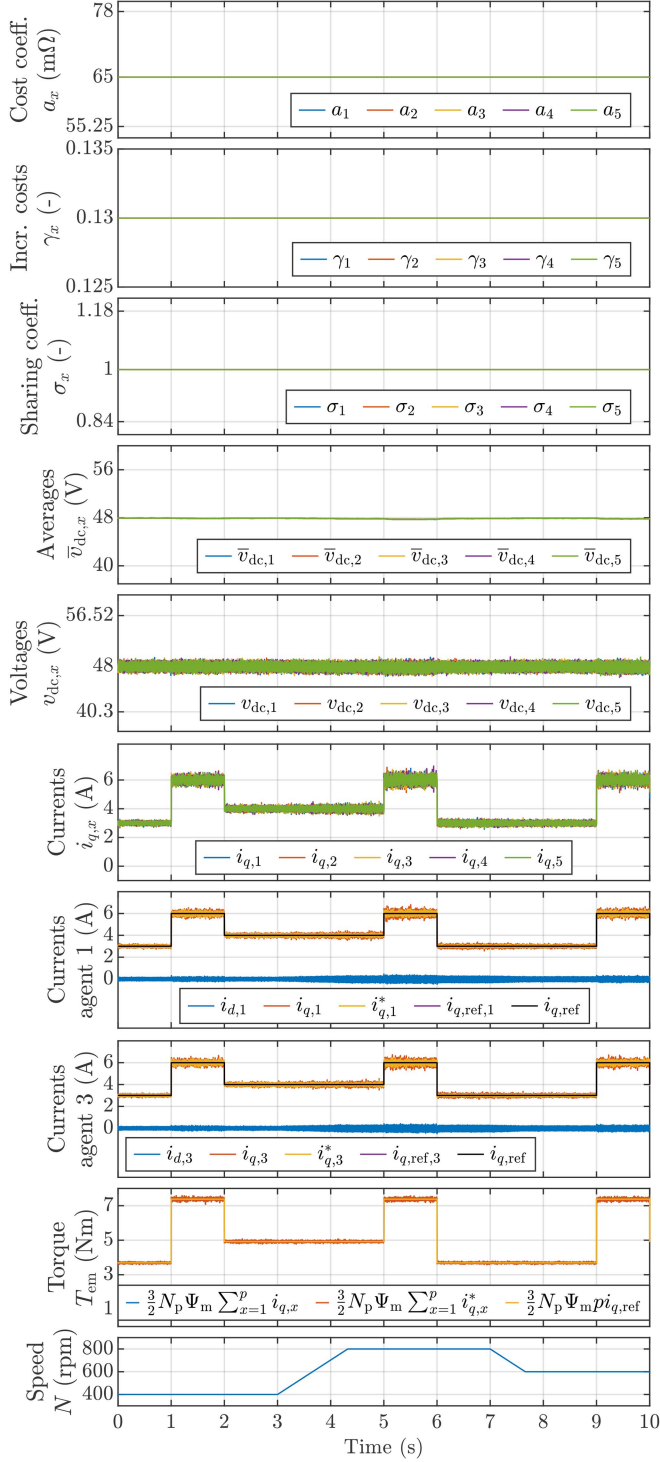


Fig. 6. Simulation results for constant, equal cost coefficients $a_x = R_s$.

still constant in time. The deterministic static average consensus algorithm (11) reaches agreement on an incremental cost $\gamma_x = 0.1307$ for all the agents x in the multi-three-phase drive, and determines the accompanying agent-specific sharing coefficients σ_x by means of (13). Agents with a higher cost coefficient a_x receive a lower sharing coefficient σ_x , while agents with a lower a_x receive a higher σ_x . The dynamic

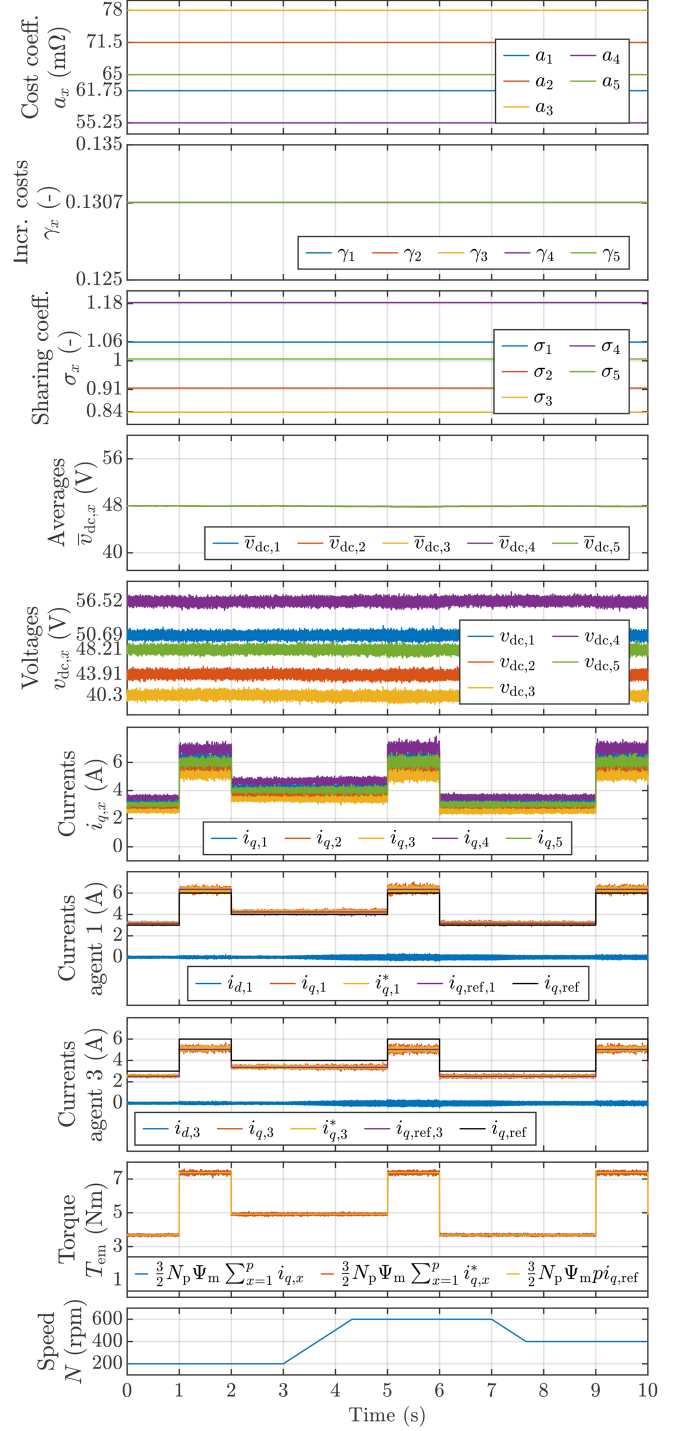


Fig. 7. Simulations results for constant, agent-specific cost coefficients a_x .

average consensus algorithm (16) still tracks the average capacitor voltage $\bar{v}_{dc,x} \approx \sum_{y=1}^p v_{dc,y}/p \approx E_{dc}/p = 48$ V, while the individual capacitor voltages $v_{dc,x}$ are correctly regulated towards their agent-specific voltage set-points $v_{dc,x}^* = \sigma_x \bar{v}_{dc,x}$ by means of the voltage balancers. The decentralized current controllers make sure that the stator current components $i_{d,x}$ and $i_{q,x}$ follow their set-points, and that the demanded torque is delivered at any time.

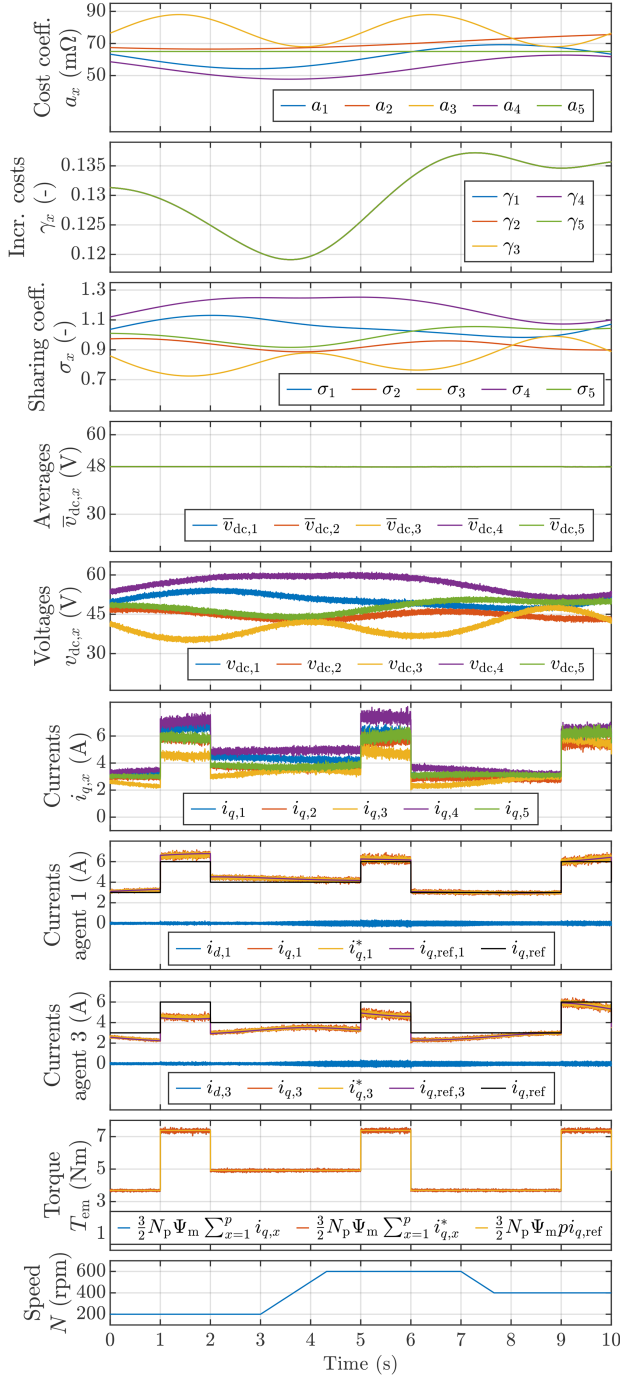


Fig. 8. Simulations results for time-varying, agent-specific cost coefficients a_x .

The simulation results of Fig. 8 show that the multi-agent control can also deal with time-varying cost functions. This situation can, for instance, occur when changes in the stator resistances $R_{s,x}$ arise due to temperature variations, and the real-time values of $R_{s,x}$ are estimated online. The agents now agree on a time-varying incremental cost γ_x by means of their deterministic static average consensus algorithm (11), resulting in agent-specific, time-varying sharing coefficients σ_x , and hence also agent-specific time-varying capacitor voltages $v_{dc,x}$ and q -current components $i_{q,x}$. However, the average capacitor

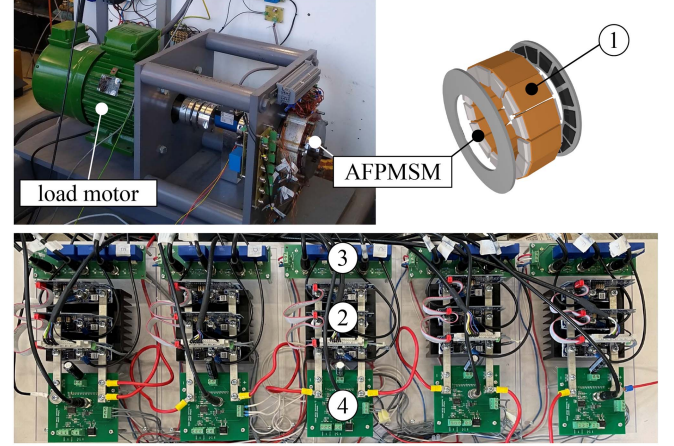


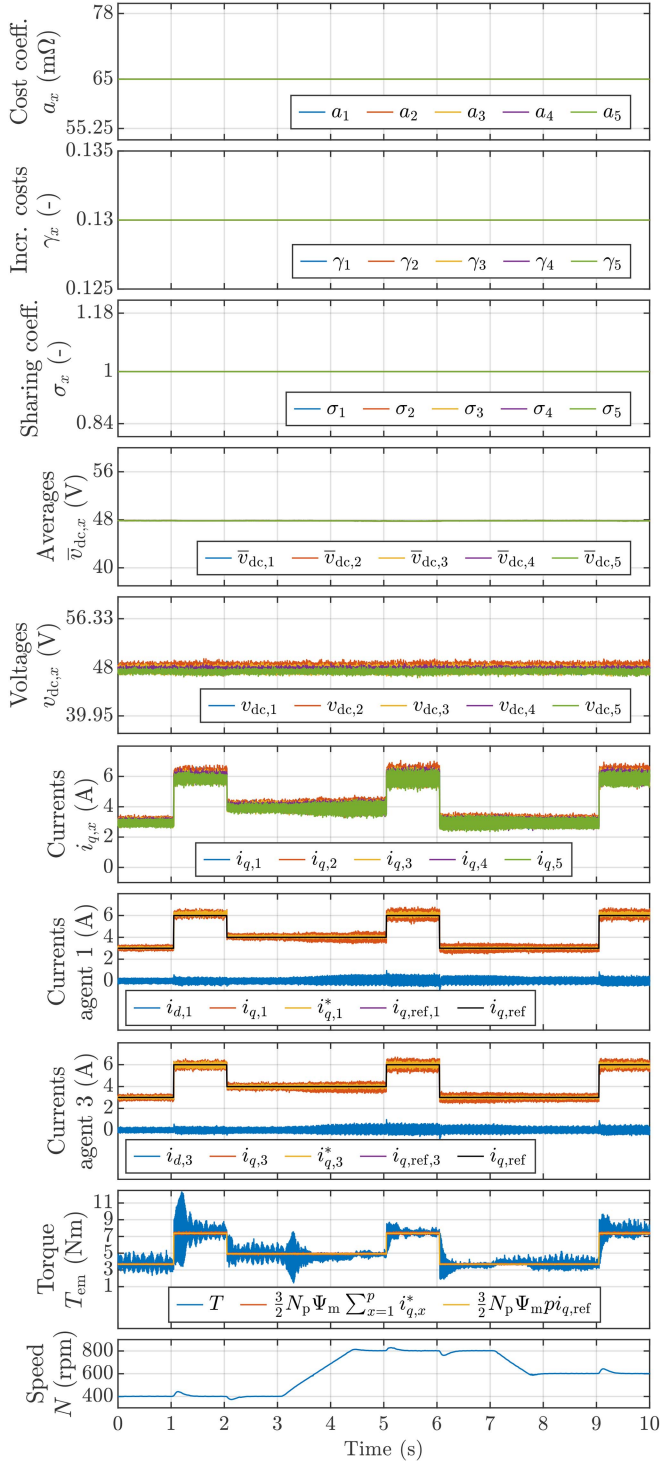
Fig. 9. In the test setup, the AFPMSM is mounted on the same shaft as a load motor. The AFPMSM consists of fifteen stator coils (1), which are divided into five three-phase agents with a separate neutral point. Each three-phase winding set is equipped with its own three-phase 2L-VSI (2). The 2L-VSIs are connected in series to the same dc power source. Each agent receives feedback from its own three current sensors (3), and its own voltage sensor (4).

voltage $\bar{v}_{dc,x}$ remains constant in time, since E_{dc} does not change in time. The torque demand T_{ref} is always met.

B. Experimental Validation

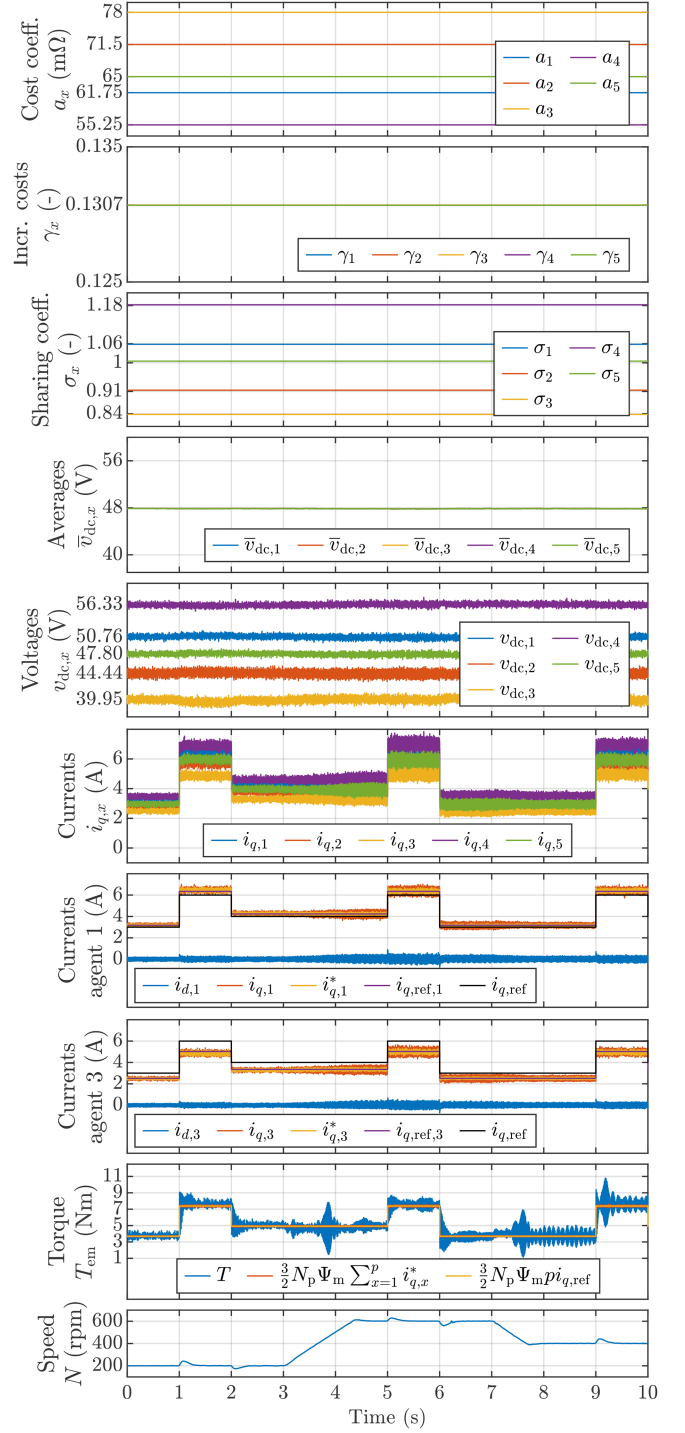
1) *Test Setup*: The test setup that is used to validate the multi-agent control strategy, is presented in Fig. 9. It consists of a multi-three-phase AFPMSM working in motoring mode, mounted on the same shaft as an induction motor. The AFPMSM has surface-mounted permanent magnets, and its specifications are summarized in Table I. It consists of fifteen stator cores with concentrated windings, which are split in five three-phase winding sets with separated neutral points. Each three-phase winding set is equipped with a dedicated low voltage 2L-VSI from Infineon, a dc-link capacitor of 220 μ F, three LEM LA 25-P current transducers, and a voltage sensor based on Texas Instrument's AMC3301 isolated amplifier, in order to form a so-called agent. The agents are connected in series to a single dc power source, which provides a total dc-bus voltage E_{dc} of 240 V. The induction load motor is driven by means of a Siemens SINAMICS drive and determines the speed of the test setup. The torque is measured with a Lorenz DR-2112 torque transducer, which is mounted on the shaft between the AFPMSM and the induction motor.

Although the multi-agent control scheme (with the control parameters of Table II) consists of five separate controllers with only neighbor-to-neighbor communication as indicated in Fig. 2, all five controllers are implemented on the same FPGA of a dSPACE MicroLabBox, to facilitate the simultaneous data acquisition for all five agents. Nevertheless, the communication frequencies $f_c = 2$ kHz and $f_{vb} = 1$ kHz are chosen within the range of state-of-the-art communication protocols [25]. A communication link between agents x and y has to exchange the variables γ_x , γ_y , e_x , and e_y at a frequency f_c of 2 kHz, and the variables p_x , p_y , $\bar{v}_{dc,x}$, and $\bar{v}_{dc,y}$ at a frequency f_{vb} of 1 kHz. When each variable is represented by means of


 Fig. 10. Experimental results for constant, equal cost coefficients $a_x = R_s$.

16 bits, this means that $4 \times 16 \times 2 \cdot 10^3 + 4 \times 16 \times 1 \cdot 10^3 = 192 \cdot 10^3$ bits must be transmitted per second. This is well below the feasible bit rate of, for instance, SPI communication, which can achieve bit rates of 50 Mbps [26].

The measured variables shown in Figs. 10–12 are obtained from the same sensors that are providing feedback to the controllers implemented on the MicroLabBox. They are


 Fig. 11. Experimental results for constant, agent-specific cost coefficients a_x .

recorded with a sampling rate of 10 kHz using ControlDesk software.

2) *Test Procedure:* The multi-agent control algorithm is validated by adhering to the following test procedure:

- 1) Validate the decentralized current control of Section IV-D when the agents are connected in parallel to a dc power source with a dc-bus voltage equal to $E_{dc}/p = 48$ V. All the agents directly receive their current set-points $i_{d,x}^*$ and

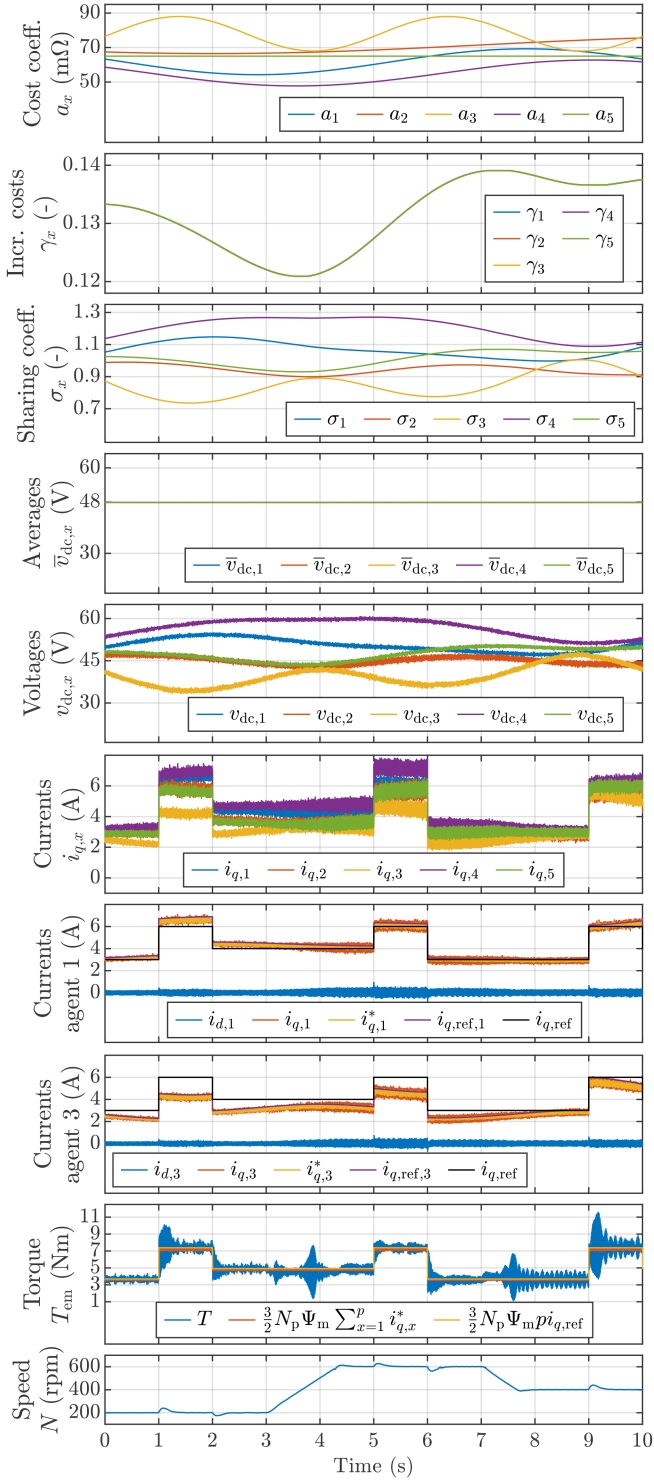


Fig. 12. Experimental results for time-varying, agent-specific cost coefficients a_x .

$i_{q,x}^*$. Neither the average consensus algorithms, nor the voltage balancers are included yet.

- 2) Add the deterministic static average consensus algorithm of Section IV-A to the decentralized current control. The agents are still connected in parallel. The sharing coefficients σ_x are directly used to determine the

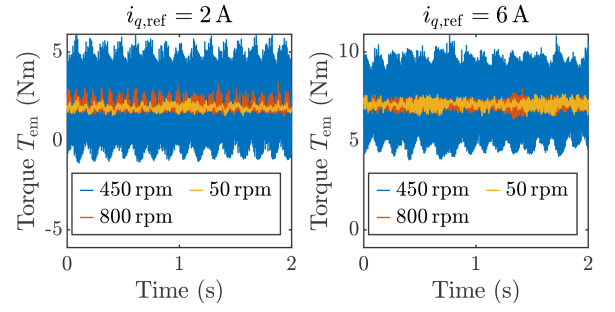


Fig. 13. Measured torque ripple at different speeds and torque set-points.

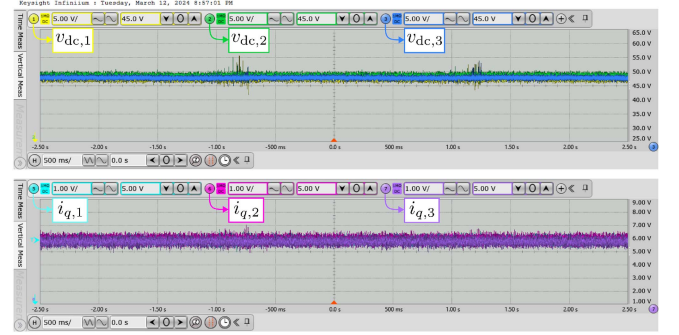


Fig. 14. Oscilloscope waveforms for constant, equal cost coefficients $a_x = R_s$ ($i_{q,ref} = 5$ A, $N = 500$ rpm).

current set-points $i_{d,x}^* = i_{d,ref,x}$ and $i_{q,x}^* = i_{q,ref,x}$. Run the dynamic average consensus algorithm in the background to check its performance, but do not activate the voltage balancers yet.

- 3) Connect the agents in series to validate the full-fledged algorithm of Fig. 2 (including static and dynamic average consensus, voltage balancing, and decentralized current control). Start with a low dc-bus voltage E_{dc} , and gradually increase it.

3) *Experimental Results:* Figs. 10-12 present the experimental validation of the proposed multi-agent control strategy of Fig. 2 for respectively equal, agent-specific constant and time-varying cost coefficients a_x . They are in good agreement with the simulation results of Figs. 6-8. A large a_x is accompanied by a low capacitor voltage $v_{dc,x}$ and q -current component $i_{q,x}$, while a small a_x corresponds to a high $v_{dc,x}$ and $i_{q,x}$. However, even when all a_x are equal – which is the case in Fig. 10 – there are small discrepancies between the currents and voltages of the agents. These are caused by small asymmetries between the real agents in the test setup. These small asymmetries are due to, for instance, manufacturing tolerances in the machine windings, the 2L-VSIs, the sensors, etc. The torque steps also cause small speed disturbances. The large ripples in the measured torque T around a speed of 450 rpm are caused by resonances in the drive. Fig. 13 confirms that the torque ripple has a large amplitude around this mechanical speed.

The validity of the measurement results of Figs. 10-12 is supported by the oscilloscope waveforms of Figs. 14-16. They are recorded with a Keysight EXR058 A oscilloscope (500 MHz,

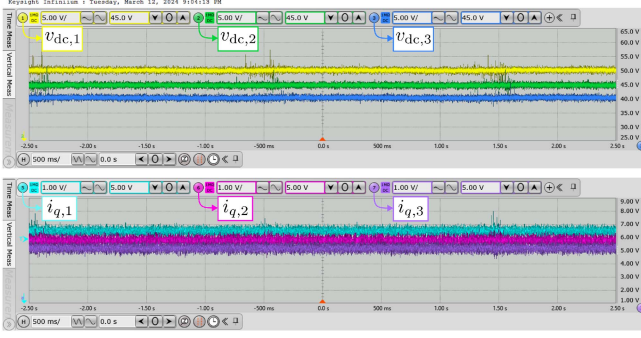


Fig. 15. Oscilloscope waveforms for constant, agent-specific cost coefficients a_x ($i_{q,\text{ref}} = 5$ A, $N = 500$ rpm).

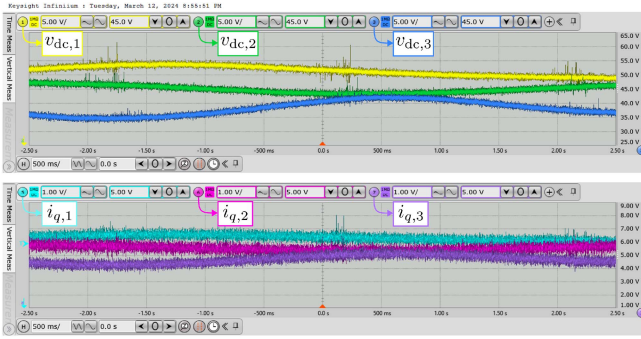


Fig. 16. Oscilloscope waveforms for time-varying, agent-specific cost coefficients a_x ($i_{q,\text{ref}} = 5$ A, $N = 500$ rpm).

50 kSa/s). The currents $i_{q,1}$, $i_{q,2}$ and $i_{q,3}$ are obtained by means of the 200 kHz LEM LA 25-P current transducers, which also provide feedback to the controllers at a sampling rate of 10 kHz. The dc-link capacitor voltages $v_{dc,1}$, $v_{dc,2}$ and $v_{dc,3}$ are measured with 150 MHz Micsig MDP3001 differential probes. The experimental results hence confirm the feasibility of the decentralized control scheme: the agents can indeed divide the torque among them in an appropriate, and (if required) unequal way, while maintaining a stable distribution of the total dc-bus voltage.

VI. COMPARISON WITH STATE-OF-THE-ART

The multi-agent voltage balancing algorithm of [11] is also able to keep the dc-link voltages of the agents stable by making use of local measurements, local computations, and local agent interactions only. For this purpose, the same dynamic average consensus algorithm (16) and the same voltage balancers (21) are used as in this work. However, in [11], it is explicitly assumed that all the agents are identical. There is no option to take into account the present condition of each agent by means of agent-specific cost functions, and static average consensus algorithm (11) is omitted. The torque and total dc-link voltage are hence always equally distributed among the agents. More specifically, this means that all the sharing coefficients σ_x are set equal to 1 a priori. All the current references $i_{d,\text{ref},x}$ and $i_{q,\text{ref},x}$ are hence equal to the global current references $i_{d,\text{ref}}$

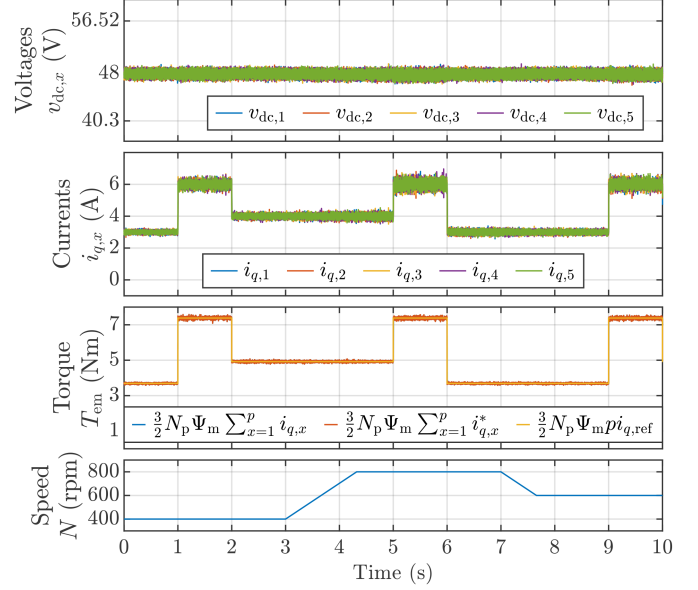


Fig. 17. Simulation results for the simplified multi-agent voltage balancing algorithm of [11].

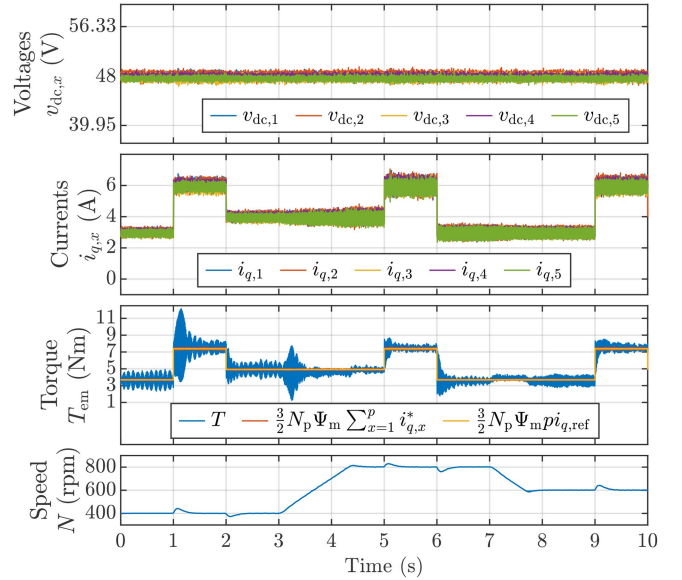


Fig. 18. Experimental results for the simplified multi-agent voltage balancing algorithm of [11].

and $i_{q,\text{ref}}$, respectively, which are derived from the global torque reference $t_{\text{ref}} = T_{\text{ref}}/p$. The dynamic average consensus algorithm (16) immediately outputs the voltage set-points for the voltage balancers, i.e. $v_{dc,x}^* = \bar{v}_{dc,x}$. Figs. 17 and 18 validate that the simplified algorithm of [11] is indeed able to ensure a stable, equal distribution of the torque and the total dc-link voltage among the agents. The small discrepancies between the agents in the experimental results of Fig. 18 are again caused by small asymmetries in the test setup. In order to obtain these simulation and measurement results – which are almost identical to the results of Figs. 6 and 10 – the algorithm in [11] requires less

computational resources and less communication load than the algorithm proposed in this paper (due to the omission of static average consensus algorithm (11) in [11]). However, in [11], there is no possibility to distribute the torque and total dc-link voltage unequally among the agents if they are not identical. The results of Figs. 7, 8, 11 and 12 can hence not be obtained by means of the simplified multi-agent voltage balancing algorithm of [11].

VII. CONCLUSION

In this work, a multi-agent, decentralized control scheme was introduced for uneven power distribution in a multi-three-phase motor drive with a stacked polyphase bridges converter. The multi-three-phase drive was composed of multiple so-called agents, each consisting of a three-phase winding set with a separate neutral point, a dedicated 2L-VSI, and a dedicated controller. The 2L-VSIs were connected in series to the same dc voltage source. The agents were not required to be identical. Instead, each agent was provided with its own cost function, reflecting its agent-specific present condition. A deterministic static average consensus algorithm making use of these cost functions allowed the agents to reach agreement on so-called sharing coefficients to distribute the total required power among them in a fair, but not necessarily even way. A dynamic average consensus algorithm in combination with decentralized voltage balancers ensured a corresponding, stable distribution of the total dc-bus voltage over the agents. The multi-agent control strategy was completed by means of decentralized PI current controllers, which regulated the currents through the agents to their agent-specific current set-points. Simulations and experiments confirmed that the current and voltage distribution was indeed stable and in accordance with the present condition of each agent, and that no central controller was required for this purpose.

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